OIML R 76-2:2007 PATTERN EVALUATION TEST REPORT

Non-Automatic Weighing Instrument (NAWI) Metrological Verification

Regional Reference Standard Laboratory (RRSL) — Ministry of Consumer Affairs, Food & Public Distribution

Session ID: SESS-2026-XP-15K_Industrial_Bench
Report Revision: Revision #1
Status: DRAFT
Date: 2026-09-28 17:17 UTC

MANUFACTURER **Mettler Toledo / Essae**	MODEL / SERIAL NUMBER **XP-15K Industrial Bench / SN-2026-IND-8849**	ACCURACY CLASS **Class III**
MAX / MIN CAPACITY **15 kg / 0.1 kg**	INTERVAL (E / D) **e = 0.005 kg / d = 0.005 kg**	ZERO-ERROR (E0) **0 kg**
RULE ENGINE VERSION **OIML-R76-1:2006-INITIAL-VERIFICATION-v1.0**	LRS SHA-256 HASH **098038205a29b331...**	REVIEWER / INSPECTOR ID **Unassigned Reviewer**

OVERALL INSTRUMENT VERDICT: FAILED

PROTOTYPE DRAFT

FORM 1: WEIGHING PERFORMANCE TEST (CLAUSE 3.5.1 & TABLE 6)

STEP	UNROUNDED IND. (P)	RAW ERROR (E)	CORRECTED (EC)	MPE LIMIT	VERDICT	GOVERNING RANGE
1	100.0025 kg	0.0025 kg	**0.0025 kg**	±0.007500 kg	PASS	2,000e < m <= 10,000e
2	2500.0025 kg	0.0025 kg	**0.0025 kg**	±0.007500 kg	PASS	2,000e < m <= 10,000e
3	5001.0025 kg	1.0025 kg	**1.0025 kg**	±0.007500 kg	FAIL	2,000e < m <= 10,000e
4	7501.0025 kg	1.0025 kg	**1.0025 kg**	±0.007500 kg	FAIL	2,000e < m <= 10,000e
5	10002.0025 kg	2.0025 kg	**2.0025 kg**	±0.007500 kg	FAIL	2,000e < m <= 10,000e
6	12502.0025 kg	2.0025 kg	**2.0025 kg**	±0.007500 kg	FAIL	2,000e < m <= 10,000e
7	15003.0025 kg	3.0025 kg	**3.0025 kg**	±0.007500 kg	FAIL	2,000e < m <= 10,000e

DIAGNOSTIC HYSTERESIS ANALYSIS (|EC,DESC - EC,ASC| ≤ |MPE|)

TEST LOAD	ASCENDING EC	DESCENDING EC	HYSTERESIS DIFFERENCE	REFERENCE FORMULA
No ascending/descending load pairs recorded yet				

FORM 3: CORNER ECCENTRICITY TEST (CLAUSE 3.6.2 & A.4.7 — LOAD = 1/3 MAX = 5 KG)

LOADING POSITION	LOAD	INDICATION	UNROUNDED (P)	CORRECTED (EC)	MPE LIMIT	VERDICT
Center	*Observation Pending*					
Front-Left	*Observation Pending*					
Back-Left	*Observation Pending*					
Back-Right	*Observation Pending*					
Front-Right	*Observation Pending*					

FORM 5: REPEATABILITY TEST (CLAUSE 3.6.1 & A.4.10 — SPREAD ≤ |MPE|)

SERIES BRACKET	INDICATIONS SAMPLE	MIN	MAX	SPREAD	MPE	VERDICT
50% Max	*Series Pending*					
100% Max	*Series Pending*					

FORM 15: EXAMINATION OF SOFTWARE-CONTROLLED WEIGHING INSTRUMENTS (CLAUSE 5.5 & WELMEC 7.2)

ITEM	REGULATORY STANDARD	MANDATORY EXAMINATION CRITERIA	EVALUATION
Software examination pending review			

OIML R 76-2 CRYPTOGRAPHIC TRACEABILITY & ED25519 DIGITAL SEAL

Certificate ID: `OIML-RRSL-SESS-2026-XP-15K_Industrial_Bench-R1`

Ed25519 Public Key Fingerprint:
`SHA256:9f1522395e39d0c669ce7a593641b35a5ec4db9f66006623d5f606d98cec974d`

Legally Relevant Software (LRS): SHA-256: `098038205a29b33152d2ef32...` (CRC-32: N/A)

Ed25519 Digital Signature: `KNU3G378MHv2AvEY4FIjAXCN0kI7...uqtv_X_4jAVdqgCg`

Reviewer Remarks: Draft report: pending formal review and digital sign-off.

Offline Verification: Cryptographically verifiable 100% offline via Ed25519 (RFC 8032). Inspectors can scan the QR code with any standard reader or offline metrology terminal to verify authenticity without internet connectivity.

PROTOTYPE
DRAFT